DIPLOMA PROGRAMME PRACTICE · HL · SET A

Biology / Paper 2

Worked mark scheme · 80 marks · 150 minutes

Award one mark per marking point. Credit equivalent reasoning and suitable units. Multiple-choice decisions carry one mark each. Answer all of Section A and two of three Section B questions. Each extended response carries 15 content marks and one mark for overall quality.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 1 [12]

Heat exposure and a heritable shell trait

(a) Calculate the survival reduction caused by heat for each morph in percentage points and compare the reductions. [3]

1 Light survival falls from 85% to 72%, a reduction of 13 points.

1 Dark survival falls from 88% to 48%, a reduction of 40 points.

1 The observed reduction is 27 points greater in the dark morph.

(b) Assuming equal starting numbers, determine the light-morph frequency among hot-chamber survivors and explain why it is not an allele frequency. [3]

1 Light survivors=72 out of 120.

1 Their frequency is 0.60 or 60%.

1 Phenotypes may represent more than one genotype and the inheritance relationship is unspecified.

(c) Explain why dividing siblings between treatments strengthens a temperature comparison and why individual snails do not all represent independent temperature replicates. [3]

1 Family background is represented in both conditions, reducing confounding between genotype or rearing history and treatment.

1 Snails sharing a chamber experience shared environmental variation.

1 Independent chambers, with chamber-level survival estimates, provide replication of the temperature treatment.

(d) Evaluate the claim that the next generation in the wild must have 60% light individuals. [3]

1 Survival is only one component of fitness and surviving snails may differ in reproductive success.

1 Inheritance, mating patterns and genotypes affect offspring phenotypes.

1 Field temperature, predators and migration may differ from chamber conditions; measure offspring or field recruitment to test the prediction.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 2 [12]

Testing a mitochondrial inhibitor

(a) Calculate ATP synthesized per molecule of O_2 consumed under each condition and the percentage decrease in this ratio. [3]

1 Control ratio=96/24=4.0.

1 With compound the ratio=30/30=1.0.

1 Decrease=(4−1)/4×100=75%.

(b) Explain the relationship between electron transport and the proton gradient under control conditions. [3]

1 Electrons from reduced carriers pass through a series of inner-membrane carriers.

1 Energy released drives proton transport from matrix to intermembrane space.

1 Oxygen accepts electrons and protons at the end of the chain to form water.

(c) Explain how a proton leak could produce all three observed changes. [3]

1 Protons return to the matrix by a route bypassing ATP synthase, reducing the gradient.

1 Less proton flow is coupled to ATP production, reducing ATP formed per oxygen consumed.

1 Electron transfer and oxygen consumption can continue or increase as opposition from the gradient decreases.

(d) A second compound stops all electron transfer to oxygen. Predict two outcomes and identify one useful control when testing these compounds. [3]

1 Oxygen consumption by the respiratory chain would fall towards zero.

1 The gradient would dissipate and sustained oxidative ATP synthesis would fall.

1 Use a vehicle-only control or matched mitochondrial protein content, substrate supply and temperature to isolate compound effects.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 3 [12]

Tracking parental DNA

(a) Use strand inheritance to explain the intermediate band after one round. [3]

1 The two strands of each heavy molecule separate.

1 Each parental strand templates synthesis of a light daughter strand.

1 Each resulting duplex has one old heavy and one new light strand and therefore intermediate density.

(b) Predict the percentages following a third round and justify your prediction. [3]

1 Each hybrid yields one hybrid and one light duplex, whereas each light duplex yields two light duplexes.

1 The number of hybrids remains constant while the total molecule number doubles.

1 Expected bands are 25% intermediate and 75% light, with no fully heavy duplexes.

(c) Explain why observations from both rounds distinguish semiconservative replication from a simple dispersive model that mixes old and new segments in every daughter strand. [3]

1 After one round both models can yield intermediate-density duplexes.

1 In a simple dispersive model all molecules after two rounds should still contain some heavy DNA and have reduced intermediate density.

1 A distinct fully light band beside an intermediate band supports segregation of intact old strands.

(d) Explain why the two new strands at one replication fork are synthesized differently despite complementary base pairing on both templates. [3]

1 The parental strands are antiparallel.

1 DNA polymerase adds to a 3′ end, so both new strands elongate only 5′ to 3′.

1 One proceeds continuously toward the fork while the other forms fragments away from it that ligase joins.

SECTION A · COMPULSORY
PAPER 2 · QUESTION 4 [12]

Water conservation after exercise

(a) Calculate approximate urinary osmole excretion in mOsm h^{-1} in each group and compare it with the difference in urine volume. [3]

1 A: $0.180 \times 310 = 55.8$ mOsm h^{-1}.

1 B: $0.075 \times 640 = 48.0$ mOsm h^{-1}.

1 B produces much less volume (about 42% of A's), while osmole excretion is about 86% of A's.

(b) Explain the chain from increased plasma osmolarity to increased collecting-duct water permeability. [3]

1 Hypothalamic osmoreceptors detect an increased plasma osmolarity or water deficit.

1 ADH release from the posterior pituitary increases.

1 ADH acts on collecting-duct cells to increase aquaporin availability in their membranes.

(c) Explain how that permeability change leads to group B's urine pattern. [3]

1 The medullary interstitial fluid has a lower water potential than the tubular fluid.

1 More water moves out of the collecting duct by osmosis and returns to the blood.

1 Less water remains in urine and its solutes are more concentrated.

(d) Evaluate the claim that the drink volume alone caused the entire difference in ADH. [3]

1 The group means show an association consistent with osmoregulation.

1 Initial hydration or individual sweat losses may differ between groups.

1 Measure baselines and sweat loss and use randomized or crossover treatments with standardized recovery to isolate the drink effect.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 5 [16]

From motor neuron to shortening muscle

(a) Outline how the electrical signal initiates chemical transmission at the motor neuron terminal. [3]

1 An action potential depolarizes the presynaptic terminal.

1 Voltage-gated calcium channels open and Ca^{2+} enters.

1 Vesicles fuse with the membrane, releasing acetylcholine by exocytosis.

(b) Explain the sequence from transmitter binding to exposure of myosin-binding sites on actin. [5]

1 Acetylcholine binds receptors on the muscle membrane.

1 Ion channels open and depolarization can initiate a muscle action potential.

1 Depolarization spreads over the membrane and into the fibre via transverse tubules.

1 Calcium is released from the sarcoplasmic reticulum.

1 Calcium binds troponin, moving tropomyosin away from binding sites on actin.

(c) Explain cross-bridge cycling and how its microscopic effects produce shortening and relaxation. [7]

1 An energized myosin head attaches to exposed actin.

1 Release of phosphate and then ADP accompanies the power stroke, pulling actin toward the sarcomere centre.

1 ATP binding causes myosin to detach from actin.

1 ATP hydrolysis re-energizes the head for another cycle.

1 Repeated sliding increases overlap and draws Z lines together without shortening the filaments.

1 Removal of stimulation allows Ca^{2+} to be pumped back into the sarcoplasmic reticulum using energy.

1 As cytosolic calcium falls, binding sites are covered and active cross-bridge cycling subsides.

(Quality) Quality of the complete extended response [1]

1 Award one mark when the complete response communicates the relevant biological reasoning clearly and concisely, with logical connections and little repetition. Assess this across the parts, independently of the content score.

SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 6 [16]

The same genome, different cells

(a) Distinguish genome, transcriptome and proteome in these tissues. [3]

1 Genome refers to the genetic material or DNA sequence present.

1 Transcriptome refers to RNA transcripts present in a cell at a specified time.

1 Proteome refers to the proteins expressed under those conditions.

(b) Explain how interactions at DNA can produce different transcription rates in the two tissues. [5]

1 Different cell types contain different regulatory proteins.

1 Specific transcription factors bind regulatory DNA sequences.

1 Activators can help recruit or stabilize transcription machinery, while repressors can reduce initiation.

1 Chromatin accessibility influences the ability of factors to bind.

1 Consequently transcripts can be produced at different rates despite identical coding sequence.

(c) Explain how RNA processing and subsequent events can affect the form and abundance of functional receptor present at the cell surface. [7]

1 Pre-mRNA can contain both exons and introns.

1 Splicing removes introns and joins exons.

1 Alternative splicing retains different exon combinations and can alter the polypeptide sequence.

1 Differences in mature mRNA stability affect how long it is available for translation.

1 Ribosomes translate the mRNA and select amino acids through complementary tRNA anticodons.

1 Folding and processing of secretory-pathway proteins influence whether a functional receptor reaches the membrane.

1 Rates of receptor removal or degradation also affect abundance, so transcript levels alone need not predict the surface level.

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SECTION B · EXTENDED RESPONSE
PAPER 2 · QUESTION 7 [16]

Evolution in a changing population

(a) Outline the assumptions directly needed for random mating to produce Hardy–Weinberg genotype proportions at a two-allele autosomal locus. [3]

1 Gametes unite randomly with respect to the locus.

1 Male and female gamete pools have the same allele frequencies p and q.

1 With a sufficiently large number of random fertilizations, expected genotype frequencies approach p^2, 2pq and q^2.

(b) Explain how founding and genetic drift can change allele frequencies without a fitness difference. [5]

1 Founders are a sample of the mainland gene pool.

1 Their allele frequencies may differ from the mainland by chance.

1 Rare alleles may be absent from the founders.

1 Finite populations experience random differences in allele transmission across generations.

1 In a small population drift is stronger and can lead to fixation or loss of alleles.

(c) Explain how selection and reduced gene flow could contribute to speciation, and distinguish evidence of divergence from evidence of reproductive isolation. [7]

1 Individuals vary in heritable traits affecting predator avoidance.

1 Those leaving more surviving offspring transmit their alleles disproportionately.

1 Island allele frequencies can change directionally under selection.

1 Reduced immigration limits homogenizing exchange with the mainland.

1 Genetic and phenotypic differences can accumulate over generations.

1 Differences in appearance or allele frequency demonstrate divergence but do not by themselves establish a reproductive barrier.

1 Evidence such as assortative mating or reduced fertility of hybrids would test reproductive isolation.

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